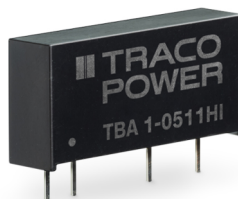


- Continuous short circuit protection
- I/O isolation: 3'000 VDC
- Operating temperature range
-40 to +85 °C without derating
- Input voltage ranges ($\pm 10\%$):
5, 12, 24 VDC
- High efficiency up to 82%
- SIP-7 package
- Unregulated outputs
- 3-year product warranty



The TBA 1HI is a 1 Watt DC/DC SIP converter series which is specifically designed to offer a low-cost solution with no concession on quality and lifetime. The new design improves on the industry standard features and offers an integrated continuous short circuit protection circuit, an operating temperature range from -40°C to 85°C without derating and I/O-isolation of 3'000 VDC. It offers a broad application range in any space and cost critical application.

Models

Order Code	Input Voltage Range	Output 1		Output 2		Efficiency typ.
		Vnom	I _{max}	Vnom	I _{max}	
TBA 1-0511HI	4.5 - 5.5 VDC (5 VDC nom.)	5 VDC	200 mA			79 %
TBA 1-0519HI		9 VDC	111 mA			80 %
TBA 1-0512HI		12 VDC	84 mA			82 %
TBA 1-0513HI		15 VDC	66 mA			82 %
TBA 1-0521HI		+5 VDC	100 mA	-5 VDC	100 mA	79 %
TBA 1-0522HI		+12 VDC	41 mA	-12 VDC	41 mA	82 %
TBA 1-0523HI		+15 VDC	33 mA	-15 VDC	33 mA	82 %
TBA 1-1211HI	10.8 - 13.2 VDC (12 VDC nom.)	5 VDC	200 mA			79 %
TBA 1-1219HI		9 VDC	111 mA			79 %
TBA 1-1212HI		12 VDC	84 mA			80 %
TBA 1-1213HI		15 VDC	66 mA			80 %
TBA 1-1221HI		+5 VDC	100 mA	-5 VDC	100 mA	79 %
TBA 1-1222HI		+12 VDC	41 mA	-12 VDC	41 mA	80 %
TBA 1-1223HI		+15 VDC	33 mA	-15 VDC	33 mA	80 %
TBA 1-2411HI	21.6 - 26.4 VDC (24 VDC nom.)	5 VDC	200 mA			79 %
TBA 1-2419HI		9 VDC	111 mA			80 %
TBA 1-2412HI		12 VDC	84 mA			82 %
TBA 1-2413HI		15 VDC	66 mA			82 %
TBA 1-2421HI		+5 VDC	100 mA	-5 VDC	100 mA	79 %
TBA 1-2422HI		+12 VDC	41 mA	-12 VDC	41 mA	82 %
TBA 1-2423HI		+15 VDC	33 mA	-15 VDC	33 mA	82 %

Input Specifications

Surge Voltage	5 Vin models: 9 VDC max. (1 s max.) 12 Vin models: 18 VDC max. (1 s max.) 24 Vin models: 30 VDC max. (1 s max.)
Recommended Input Fuse	5 Vin models: 500 mA (slow blow) 12 Vin models: 200 mA (slow blow) 24 Vin models: 100 mA (slow blow) (The need of an external fuse has to be assessed in the final application.)
Input Filter	Internal Capacitor (add. external 22 µF, ESR <0.1Ω, recommended)

Output Specifications

Voltage Set Accuracy	±3% max. (at 60% for 5VDC models) ±3% max. (at 80% for other models)	
Regulation (Unregulated)	- Input Variation (1% Vin step) - Load Variation - Voltage Balance (symmetrical load)	single output models: 1.5% max. dual output models: 1.5% max. See application note: www.tracopower.com/tba1hi-cc/ dual output models: 1% max.
Ripple and Noise	- 20 MHz Bandwidth	150 mVp-p max. 100 mVp-p typ.
Capacitive Load	- single output - dual output	5 Vout models: 2'200 µF max. 9 Vout models: 1'000 µF max. 12 Vout models: 470 µF max. 15 Vout models: 470 µF max. 5 / -5 Vout models: 2'200 / 2'200 µF max. 12 / -12 Vout models: 470 / 470 µF max. 15 / -15 Vout models: 220 / 220 µF max.
Minimum Load	See application note: www.tracopower.com/tba1hi-cc/	
Temperature Coefficient	±0.02 %/K max.	
Start-up Time	10 ms max.	
Short Circuit Protection	Continuous, Automatic recovery	

Safety Specifications

Standards	- IT / Multimedia Equipment	Designed for IEC/EN/UL 62368-1 (not certified)
Over Voltage Category	Not mains connected	

EMC Specifications

EMI (Emissions)	- Conducted Emissions - Radiated Emissions	EN 55032 class B (with external filter) EN 55032 class B (internal filter)
External filter proposal:		www.tracopower.com/tba1hi-emc-filter/

General Specifications

Relative Humidity	95% max. (non condensing)	
Temperature Ranges	- Operating Temperature - Case Temperature - Storage Temperature	-40°C to +95°C +105°C max. -55°C to +125°C
Power Derating	- High Temperature	5.0 %/K above 85°C See application note: www.tracopower.com/tba1hi-cc/
Cooling System	Natural convection (20 LFM)	
Switching Frequency	40 - 200 kHz (PWM)	
Insulation System	Functional Insulation	
Isolation Test Voltage	- Input to Output, 60 s	3'000 VDC

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

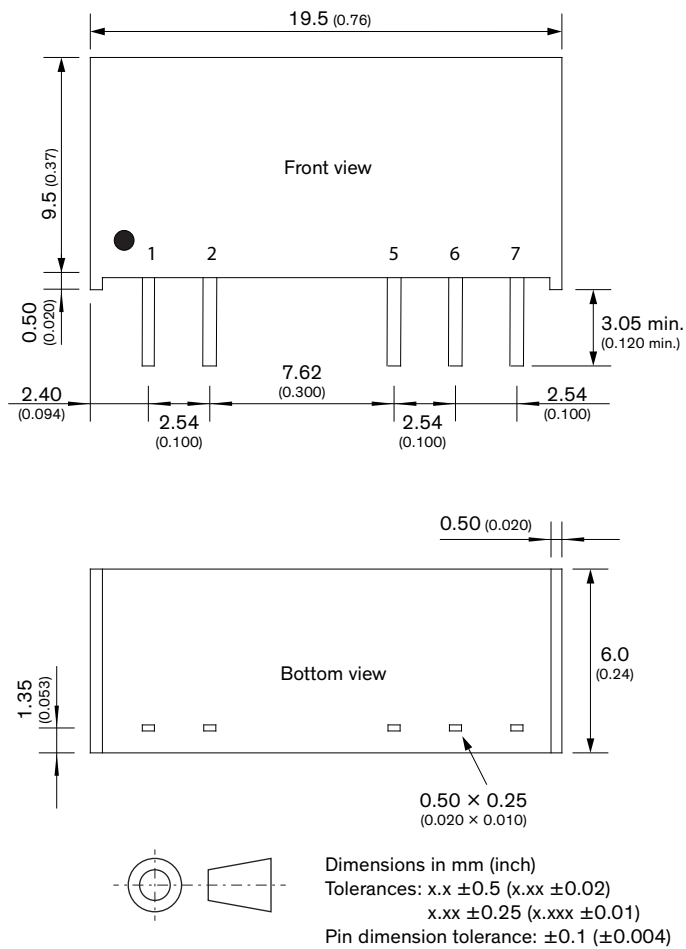
Isolation Resistance	- Input to Output, 500 VDC	1'000 M Ω min.
Isolation Capacitance	- Input to Output, 100 kHz, 1 V	10 pF typ.
Reliability	- Calculated MTBF	2'000'000 h (MIL-HDBK-217F, ground benign)
Washing Process		Not allowed
Housing Material		Plastic (UL 94 V-0 rated)
Potting Material		Epoxy (UL 94 V-0 rated)
Pin Material		Nickel-Iron (Alloy 42)
Pin Foundation Plating		Nickel (1.5 μ m min.)
Pin Surface Plating		Tin (3 μ m min.), bright
Housing Type		Plastic Case
Mounting Type		PCB Mount
Connection Type		THD (Through-Hole Device)
Footprint Type		SIP7
Soldering Profile		Lead-Free Wave Soldering 265 °C / 5 s max.
Weight		2.3 g
Environmental Compliance	- REACH Declaration	www.tracopower.com/info/reach-declaration.pdf REACH SVHC list compliant REACH Annex XVII compliant
	- RoHS Declaration	www.tracopower.com/info/rohs-declaration.pdf Exemptions: 7(a), 7(c)-I (RoHS exemptions refer to the component concentration only, not to the overall concentration in the product (O5A rule).)
	- SCIP Reference Number	838e6883-84a5-4883-9a1c-ea097f6f112b

Additional Information

Supporting Documents	www.tracopower.com/overview/tba1hi
Frequently Asked Questions	www.tracopower.com/glossary-faq
Glossary	www.tracopower.com/info/glossary.pdf
Service Portal	www.tracoserviceportal.com

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

Outline Dimensions



Pinout

Pin	Single	Dual
1	+Vin (Vcc)	+Vin (Vcc)
2	−Vin (GND)	−Vin (GND)
5	−Vout	−Vout
6	No pin	Common
7	+Vout	+Vout

Avoid routing PCB traces under the converter.